

Compound 1

Compound 2

Compounds **1** and **2** can be described as:

- ☐ A. enantiomers. [22%]
- ☐ B. diastereomers. [9%]
- ✓ ☒ C. conformational isomers. [56%]
- ☐ D. constitutional isomers. [11%]

Correct

56% answered correctly

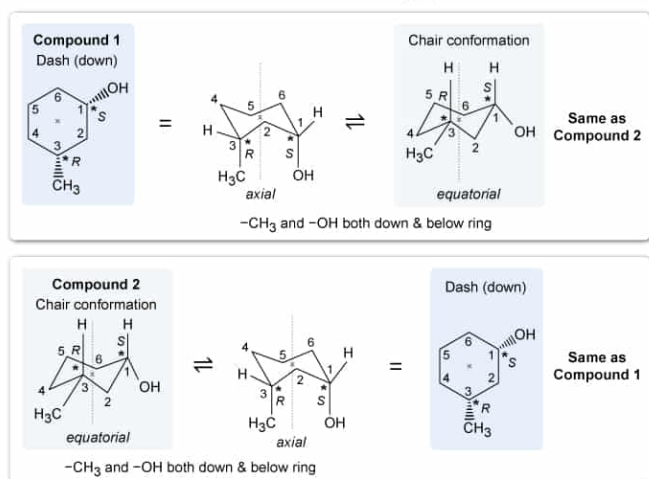
Explanation:

Conformational isomers

Identical structures that differ only in the arrangement of atoms caused simply by rotations around one or more σ bonds

* = Stereocenter

x = Ring axis



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Isomers are compounds that have the same molecular formula but a different arrangement of atoms. **Conformational isomers** are structures that have the **same connectivity** and can be interconverted by the **rotation of σ bonds**. Because conformational isomers are identical except for bond rotations, they are the **same compound**.

Six-membered rings can be depicted in two different ways: as a wedge-dash

projection or a **chair conformation**. In wedge-dash projections, wedges indicate the substituent is up and above the ring, and dashes indicate the substituent is down and below the ring. Substituents can be either **axial or equatorial** in a chair conformation. However, axial and equatorial do not dictate whether the substituent is up and above the ring or down and below the ring.

The wedge-dash projection of Compound **1** shows the -OH and the -CH_3 both on a dash (down). Likewise, the chair conformation of Compound **2** shows the -OH and -CH_3 as equatorial down. To represent this wedge-dash projection as a chair conformation, both substituents can be shown as axial down. A **chair flip** puts the substituents in an equatorial down position (still on the **same side** of the ring), yielding an identical structure to Compound **2**. Because these compounds have the same connectivity and can be interconverted from one to the other through bond rotations (via chair flip), they are conformational isomers.

(Choice A) **Enantiomers** are nonsuperimposable mirror images in which all stereocenters are inverted. The stereocenter configurations in Compounds **1** and **2** are identical, and therefore cannot be enantiomers.

(Choice B) **Diastereomers** are stereoisomers in which at least one, but not all, stereocenters are inverted. Because none of the stereocenters in Compounds **1** and **2** are inverted, they cannot be diastereomers.

(Choice D) Compounds **1** and **2** have the same molecular formula *and* the same connectivity, so they are not **constitutional isomers**.

Educational objective:

Conformational isomers are structures that have the same formula and connectivity, and can be interconverted by the rotation of σ bonds. Because conformational isomers are identical except for structural bond rotations, they are the same compound.

Time spent: 0 sec

Subject:

Organic Chemistry

QID: 401888

System:

Introduction to Organic Chemistry